

H524 Introduction to Biostatistics

Final Group Project Specification

Fall 2025

Project Timeline

Week 7-8: Project preparation homework (dataset selection, proposal)

Weeks 9-10: Data analysis and report writing

Week 10: In-class presentations (10-15 minutes per group)

End of Term: Final written report and code submission

Weight: 25% of final course grade

1 Project Overview

1.1 Learning Objectives

By completing this group project, you will:

- Apply statistical methods learned throughout the course to a real-world health dataset
- Formulate and test meaningful research questions using appropriate statistical techniques
- Create publication-quality tables, figures, and reports
- Collaborate effectively in a team environment
- Communicate statistical findings to both technical and non-technical audiences
- **Integrate AI tools responsibly to enhance your analysis while maintaining statistical rigor**
- Develop reproducible research skills using R and RMarkdown

1.2 Project Scope

This is a **GROUP project** with 4-5 students per team. Each group will:

1. Select a publicly available health-related dataset (approved in Week 7-8)
2. Develop 2-3 research questions that can be addressed using methods from this course
3. Conduct a complete statistical analysis including descriptive statistics, hypothesis testing, and regression
4. Produce a professional written report (10-15 pages)
5. Deliver an in-class presentation (10-15 minutes)
6. Submit well-documented, reproducible R code

2 Required Deliverables

2.1 Deliverable 1: Written Report (10-15 pages)

Your report should follow the structure of a scientific research article. Use RMarkdown to create a professional PDF document.

2.1.1 Executive Summary (1 page)

- Brief background and context
- Key research questions
- Main findings (2-3 bullet points)
- Primary conclusions and implications

2.1.2 Introduction & Background (2 pages)

- **Research Context:** Why is this topic important for public health?
- **Literature Review:** Summarize 3-5 relevant published studies (use PubMed, Google Scholar)
- **Research Questions:** State 2-3 specific, testable research questions
- **Hypotheses:** For each research question, state your null and alternative hypotheses
- **References:** Use APA format for all citations

2.1.3 Data & Methods (2-3 pages)

- **Dataset Description:**
 - Source and citation
 - Study design (cross-sectional, cohort, etc.)
 - Sample size and time period
 - Population characteristics
- **Variables and Measures:**
 - Primary outcome variable(s)
 - Primary exposure/predictor variable(s)
 - Confounding or effect modification variables
 - Variable types (continuous, categorical, ordinal)
 - Any transformations or recoding performed
- **Statistical Methods:**
 - Descriptive statistics used
 - Hypothesis tests performed (with justification)
 - Regression models fitted

- Significance level (α) used
- How missing data was handled

- **Software:**

- R version and key packages used
- [AI tools used and their role in the analysis](#)

2.1.4 Results (3-4 pages)

- **Descriptive Statistics:**

- Table 1: Participant characteristics (demographics, key variables)
- Summary statistics: means, SDs, medians, IQRs, frequencies, percentages
- Visual summaries: histograms, boxplots, bar charts

- **Primary Analyses:**

- Present results for each research question
- Include test statistics, p-values, and confidence intervals
- Provide effect sizes where appropriate
- Use tables and figures to communicate findings clearly

- **Table and Figure Requirements:**

- All tables numbered (Table 1, Table 2, etc.) with descriptive captions
- All figures numbered (Figure 1, Figure 2, etc.) with descriptive captions
- Tables: Use publication-quality formatting (no vertical lines, minimal horizontal lines)
- Figures: Clear axis labels, legends, and titles; readable font sizes
- Reference all tables/figures in the text

- **Statistical Reporting Standards:**

- Report exact p-values (not just “ $p < 0.05$ ”) when $p < 0.001$
- Include 95% confidence intervals for key estimates
- Report effect sizes (e.g., mean differences, odds ratios, correlation coefficients)

2.1.5 Discussion & Conclusions (2-3 pages)

- **Interpretation of Findings:**

- What do your results mean in the context of your research questions?
- How do your findings compare to the literature?
- What are the public health implications?

- **Limitations:**

- Study design limitations

- Data quality issues
- Statistical limitations
- Generalizability concerns

- **Strengths:**

- What makes this analysis valuable?
- What did you do particularly well?

- **Future Directions:**

- What additional analyses could be done?
- What data would help answer remaining questions?
- What interventions or policies might be informed by this work?

- **Conclusions:**

- Concise summary of key findings
- Main take-home messages

2.1.6 References

- Minimum of 5 references in APA format
- Include dataset citation
- Use primary sources (journal articles, government reports)

2.1.7 Appendix (if needed)

- Supplementary tables or figures
- Additional sensitivity analyses
- Extended technical details

2.2 Deliverable 2: R Code (Well-Documented)

Submit your complete R code as either:

- An RMarkdown (.Rmd) file that generates your report, OR
- A standalone R script (.R) file with all analyses

2.2.1 Code Requirements

- **Reproducibility:**

- Code runs from start to finish without errors
- Includes code to load the dataset (with instructions on where to obtain it)
- Sets random seed if any randomization is used

- Includes `sessionInfo()` at the end
- **Organization:**
 - Clear sections with headers (e.g., “Data Loading”, “Cleaning”, “Analysis”, “Figures”)
 - Logical flow from data import to final results
 - All packages loaded at the beginning
- **Documentation:**
 - Comments explaining what each section does
 - Comments for any complex or non-obvious code
 - Variable names that are clear and descriptive
- **Good Coding Practices:**
 - Consistent indentation and spacing
 - No hard-coded values that should be variables
 - Functions used for repeated operations
 - Avoid unnecessary code repetition
- **AI-Generated Code:**
 - [Mark which sections were AI-generated \(use comments\)](#)
 - [Document how you verified AI code works correctly](#)
 - [Explain any modifications you made to AI output](#)

2.3 Deliverable 3: Presentation (10-15 minutes)

Each group will present their project during Week 10.

2.3.1 Presentation Structure

- **Introduction (2-3 minutes):**
 - Research question and motivation
 - Brief background
- **Methods (2-3 minutes):**
 - Dataset overview
 - Key variables
 - Statistical methods used
- **Results (4-6 minutes):**
 - 2-3 key tables or figures
 - Main statistical findings
 - Focus on most important results

- **Discussion & Conclusions (2-3 minutes):**
 - What do findings mean?
 - Limitations
 - Take-home message
- **Q&A (5 minutes):**
 - Be prepared to answer questions about methods, results, and interpretations

2.3.2 Presentation Requirements

- Create slides using Beamer (LaTeX) or PowerPoint
- Each team member must present at least one section
- Include a title slide with all team member names
- Use clear, readable fonts (minimum 20pt for body text)
- Limit text on slides (use bullet points, not paragraphs)
- Include visual elements (tables, figures, diagrams)
- Practice timing to stay within 10-15 minutes
- Upload slides to Canvas before your presentation

3 Statistical Analysis Requirements

Your project must include **at least 4** of the following statistical methods:

1. Descriptive Statistics and Visualizations

- Summary statistics for all key variables
- At least 3 high-quality figures (histograms, boxplots, scatterplots, etc.)
- At least 1 well-formatted table of descriptive statistics

2. Confidence Intervals

- 95% CIs for means, proportions, or regression coefficients
- Proper interpretation of confidence intervals

3. Hypothesis Testing

- Choose appropriate test(s): two-sample t-test, paired t-test, one-way ANOVA, chi-square test
- State null and alternative hypotheses
- Report test statistics, p-values, and conclusions

4. Correlation Analysis

- Pearson or Spearman correlation
- Correlation matrix if multiple variables
- Scatterplots with correlation coefficients

5. Simple Linear Regression

- Fit at least one regression model
- Interpret slope and intercept
- Report R-squared
- Check model assumptions (residual plots)
- Predict values for new observations

6. Additional Methods (as appropriate)

- Stratified analyses
- Effect modification or interaction assessment
- Sensitivity analyses
- Non-parametric methods (if assumptions violated)

Note: Choose methods appropriate for your research questions and data. Quality of analysis is more important than quantity.

4 AI Integration Requirements

This course embraces AI as a tool for learning and productivity. Your project must document AI use.

4.1 Required AI Documentation

In a dedicated section of your report or in code comments, address:

1. AI Tools Used:

- Which AI tools did you use? (e.g., Microsoft Copilot, GitHub Copilot, ChatGPT)
- For what tasks? (e.g., code generation, debugging, explanation of concepts, literature search)

2. How AI Assisted Your Work:

- Provide specific examples of AI prompts and outputs
- Explain how AI helped you solve problems or understand concepts
- What percentage of your code was AI-generated vs. written by you?

3. Verification of AI Outputs:

- How did you verify AI-generated code was correct?
- Did you find any errors in AI outputs? How did you fix them?

- What manual checks did you perform?

4. Reflection on AI's Role:

- What did AI do well?
- What did AI struggle with?
- How did using AI affect your learning?
- Would you use AI differently next time?

4.2 AI Best Practices

AI Use Guidelines

Encouraged:

- + Using AI to generate starter code templates
- + Asking AI to explain statistical concepts or error messages
- + Using AI to help debug code
- + Asking AI to suggest visualization options
- + Using AI to improve code documentation

Required:

- ✓ Verify all AI outputs are correct
- ✓ Understand every line of code you submit
- ✓ Document AI use transparently
- ✓ Test AI-generated code thoroughly

Prohibited:

- Submitting AI-generated work you don't understand
- Using AI to fabricate data or results
- Copying AI output without verification
- Failing to disclose AI use when required

5 Grading Rubric

Total: 100 points = 25% of course grade

Component	Points	Criteria
Introduction & Background	10	Clear research questions, relevant literature review, well-motivated hypotheses
Methods Description	10	Complete description of data, variables, and statistical methods; appropriate method selection
Results Presentation	15	Clear tables and figures, appropriate statistical reporting, comprehensive results
Statistical Correctness	15	Correct application of methods, accurate calculations, proper interpretation
Discussion & Conclusions	10	Thoughtful interpretation, acknowledgment of limitations, clear conclusions
Component	Points	Criteria
Reproducibility	10	Code runs without errors, produces all reported results, includes session info
Code Quality & Documentation	10	Well-organized, clearly commented, follows good practices, AI use documented

5.1 Written Report (60 points)

5.2 R Code (20 points)

5.3 Presentation (15 points)

Component	Points	Criteria
Content & Clarity	10	Clear communication, effective visuals, appropriate content selection, timing
Team Participation	5	All members participate, balanced contributions, ability to answer questions

5.4 Peer Evaluation (5 points)

Note: Peer evaluation forms will be completed confidentially by each team member after the project is submitted.

6 Group Dynamics

6.1 Team Formation

- Groups of 4-5 students will be formed in Week 7
- You may self-select groups or be assigned by instructor

Component	Points	Criteria
Individual Contribution	5	Based on peer evaluations of contribution to team effort

- All group members are expected to contribute equally

6.2 Equal Contribution Expectation

Each team member should contribute to:

- Dataset selection and proposal development
- Literature review
- Data analysis and coding
- Writing different sections of the report
- Creating presentation slides
- Presenting during the oral presentation

6.3 Peer Evaluation Process

After project submission, each team member will complete a confidential peer evaluation form rating:

- Quality of contributions
- Reliability and timeliness
- Communication and collaboration
- Overall effort level

Individual grades may be adjusted up or down based on peer evaluations if there is evidence of unequal contribution.

6.4 Conflict Resolution

If conflicts arise within your group:

1. **Team Discussion:** Try to resolve issues through open communication within the group
2. **Document Issues:** Keep records of team meetings, task assignments, and communications
3. **Seek Help Early:** Contact the instructor if issues cannot be resolved internally
4. **Formal Complaint:** If needed, submit written documentation of the issue to the instructor

Waiting until the last minute to report conflicts is not acceptable. Notify the instructor as soon as problems arise.

7 Technical Specifications

7.1 File Formats

- **Written Report:** PDF (generated from RMarkdown preferred)
- **Code:** .R or .Rmd files
- **Presentation:** PDF or PowerPoint (.pptx)
- **Data:** Include dataset or clear instructions for obtaining it

7.2 File Naming Conventions

Use the following naming scheme:

- Report: `GroupX_Final_Report.pdf`
- Code: `GroupX_Analysis_Code.R` or `GroupX_Analysis_Code.Rmd`
- Presentation: `GroupX_Presentation.pdf`
- Data (if included): `GroupX_Data.csv` (or other format)

Replace “X” with your group number (e.g., Group1, Group2, etc.)

7.3 Submission Method

- Create a single ZIP file containing all required files
- Name the ZIP file: `GroupX_Final_Project.zip`
- Submit via Canvas by the deadline
- Only ONE team member needs to submit (include all names in submission)

7.4 Late Policy

- **Presentations:** Cannot be made up if missed (0 points for presentation component)
- **Written Report:** 10% penalty per day late (up to 3 days)
- **After 3 days:** No credit given
- **Extensions:** Must be requested at least 48 hours before deadline with valid reason

8 Example Projects

To help you understand expectations, here are examples of appropriate project topics and approaches:

8.1 Example 1: NHANES Physical Activity and Hypertension

Dataset: National Health and Nutrition Examination Survey (NHANES)

Research Questions:

- Is physical activity level associated with prevalence of hypertension in US adults?
- Does this association differ by age group?

Methods Used:

- Descriptive statistics for demographics and key variables
- Chi-square test for association between physical activity category and hypertension
- Two-sample t-test comparing blood pressure between active and sedentary individuals
- Logistic regression predicting hypertension from physical activity (controlling for age, BMI)
- Stratified analysis by age group

Key Findings (hypothetical):

- 32% of sedentary adults had hypertension vs. 18% of physically active adults ($p < 0.001$)
- After adjusting for age and BMI, physical activity remained protective (OR = 0.58, 95% CI: 0.45-0.75)
- Association strongest in adults over 50 years old

8.2 Example 2: Framingham Heart Study and Cholesterol

Dataset: Framingham Heart Study (teaching dataset)

Research Questions:

- What is the relationship between age and total cholesterol?
- Does smoking status affect cholesterol levels?

Methods Used:

- Descriptive statistics and histograms for cholesterol distribution
- Correlation analysis between age and cholesterol
- Simple linear regression: $\text{cholesterol} \sim \text{age}$
- Two-sample t-test comparing cholesterol between smokers and non-smokers
- Multiple linear regression: $\text{cholesterol} \sim \text{age} + \text{smoking} + \text{BMI}$

Key Findings (hypothetical):

- Moderate positive correlation between age and cholesterol ($r = 0.42$, $p < 0.001$)
- For each additional year of age, cholesterol increases by 0.85 mg/dL (95% CI: 0.70-1.00)
- Smokers had 8.2 mg/dL higher cholesterol than non-smokers ($p = 0.03$)

8.3 Example 3: COVID-19 Vaccination and Demographics

Dataset: CDC COVID-19 vaccination data (county-level)

Research Questions:

- Is county-level vaccination rate associated with median household income?
- Do urban and rural counties differ in vaccination rates?

Methods Used:

- Descriptive statistics for vaccination rates across counties
- Scatterplot and correlation between vaccination rate and income
- Simple linear regression: vaccination rate $\sim$ median income
- Two-sample t-test comparing urban vs. rural vaccination rates
- ANOVA comparing vaccination rates across regions (Northeast, South, Midwest, West)

Key Findings (hypothetical):

- Strong positive correlation between county income and vaccination rate ($r = 0.68$, $p < 0.001$)
- Each \$10,000 increase in median income associated with 5.2% higher vaccination rate
- Urban counties had 12% higher vaccination rates than rural counties ($p < 0.001$)

9 Timeline and Milestones

Week	Milestone
Week 7	Form groups; begin exploring datasets
Week 8	Submit project proposal (dataset, research questions, planned methods)
Week 9	Data cleaning and preliminary analyses; draft Methods section
Week 10 (early)	Complete analyses; draft Results section; create presentation slides
Week 10 (end)	IN-CLASS PRESENTATIONS
End of Term	Final written report and code due

9.1 Week 8 Proposal Requirements

Submit a 1-2 page proposal including:

- Group members and contact information
- Dataset selected (with citation/link)

- 2-3 proposed research questions
- List of variables to be analyzed
- Planned statistical methods
- Potential challenges or concerns

This proposal will receive feedback from the instructor to help guide your project.

10 Resources

10.1 RMarkdown Report Template

An RMarkdown template for your report is provided on Canvas:

- `H524_Project_Report_Template.Rmd`

This template includes:

- YAML header for PDF generation
- Section structure matching this specification
- Example code for tables and figures
- Citation examples

10.2 Beamer Presentation Template

A Beamer (LaTeX) template for your presentation is provided on Canvas:

- `H524_Project_Presentation_Template.tex`

This template includes OSU branding and example slides.

10.3 Peer Evaluation Form

The peer evaluation form will be distributed after presentations and is due with the final report. It includes:

- Rating scales for each team member's contributions
- Open-ended questions about team dynamics
- Confidential submission to instructor only

10.4 Dataset Resources

Recommended sources for health datasets:

- **CDC Data:** <https://data.cdc.gov/>
- **NHANES:** <https://wwwn.cdc.gov/nchs/nhanes/>
- **Framingham Heart Study:** Teaching datasets available on Canvas
- **Behavioral Risk Factor Surveillance System (BRFSS):** <https://www.cdc.gov/brfss/>
- **UCI Machine Learning Repository:** <https://archive.ics.uci.edu/ml/>
- **Kaggle Datasets:** <https://www.kaggle.com/datasets>
- **Data.gov:** <https://data.gov/>

10.5 Style Guide for Tables and Figures

Tables:

- Use horizontal lines sparingly (top, bottom, below headers)
- Avoid vertical lines
- Left-align text, right-align numbers
- Include units in column headers
- Use footnotes for clarifications
- Caption goes above the table

Figures:

- Use clear, readable fonts (minimum 10pt)
- Label all axes with units
- Include legends when multiple groups shown
- Use colorblind-friendly palettes
- Caption goes below the figure
- Export at high resolution (300 dpi for publication)

10.6 APA Citation Examples

Journal Article:

Smith, J., & Jones, M. (2023). The effect of exercise on blood pressure: A meta-analysis. *American Journal of Public Health*, 113(4), 456-467. <https://doi.org/10.2105/AJPH.2023.123456>

Dataset:

Centers for Disease Control and Prevention. (2022). *National Health and Nutrition Examination Survey 2017-2020* [Dataset]. U.S. Department of Health and Human Services. <https://www.cdc.gov/nchs/nhanes/>

Website:

World Health Organization. (2023, June 15). *Cardiovascular diseases (CVDs)*. [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))

10.7 AI Assistance

AI Tools Can Help With:

- Generating code templates for common analyses
- Debugging error messages
- Creating ggplot2 visualization code
- Formatting tables with `kable` or `gt`
- Explaining statistical concepts
- Suggesting appropriate statistical tests
- Improving code documentation
- Finding relevant literature

Remember: Always verify AI outputs and understand every line of code you submit!

11 Frequently Asked Questions

Q: Can we use data we collected ourselves?

A: Yes, with instructor approval. You must have IRB approval if human subjects data was collected for research purposes.

Q: How long should our presentation be?

A: 10-15 minutes, followed by 5 minutes for questions. Practice your timing.

Q: Can we do more than the minimum 4 statistical methods?

A: Absolutely! Quality analysis is encouraged. Just ensure you do justice to each method.

Q: What if we can't meet as a group?

A: Use Zoom, Google Docs, and other collaboration tools. Document your group meetings and task assignments.

Q: Can we change our research question after the proposal?

A: Minor changes are fine. Major changes require instructor approval.

Q: How much code documentation is enough?

A: Every analysis section should have comments. Someone unfamiliar with your project should be able to follow your code.

Q: What if a team member isn't contributing?

A: Document the issue and contact the instructor immediately. Don't wait until the end.

Q: Can we use methods not covered in class?

A: Advanced methods are welcome if you can explain them correctly. Consult with the instructor first.

Q: How do we cite AI tools?

A: Include in your methods or acknowledgments: "We used [Tool Name] [Version] to assist with [specific tasks]. All outputs were verified by the authors."

Q: Is there a page limit?

A: The report should be 10-15 pages (excluding references and appendices). Quality over quantity.

12 Academic Integrity

All work submitted must represent your team's own understanding and effort.

Academic Integrity Violations

The following constitute academic dishonesty:

- Submitting AI-generated content without verification or understanding
- Fabricating or falsifying data or results
- Plagiarizing text from sources without proper citation
- Copying code from other groups without attribution
- Having someone outside your group complete substantial portions of the project
- Submitting work completed by a previous year's group

Consequences: Academic integrity violations will result in a zero on the project and possible referral to the Office of Student Conduct.

13 Summary Checklist

Before submitting, verify you have:

- ☐ Written report (10-15 pages) with all required sections
- ☐ Executive summary, introduction, methods, results, discussion, conclusions, references
- ☐ At least 4 statistical methods appropriately applied
- ☐ Publication-quality tables and figures with captions
- ☐ R code that runs without errors and is well-documented
- ☐ AI use documented in report and code

- ☐ Presentation slides uploaded to Canvas
- ☐ All files named correctly and zipped together
- ☐ Peer evaluation form completed
- ☐ All team members' names on report and presentation
- ☐ Submitted by deadline

Good luck with your project!

Questions? Contact the instructor during office hours or via Canvas messaging.